

Caloric restriction in humans: potential pitfalls and health concerns.



To date, the only intervention that has consistently been shown to slow the rate of aging, and to increase mean and maximum lifespan in short-lived species, is life-long calorie restriction. It is yet unclear whether long-term calorie restriction in longer lived species (i.e. primates and humans) will have a similar effect. In humans, several studies investigating short-term calorie restriction or "weight loss" programs suggest beneficial outcomes on parameters of cardiovascular disease. Studies on long-term calorie restriction are performed on a self-selected group of human subjects and show similar effects. However, few studies are currently investigating the quality of life and potential pitfalls of long-term calorie restriction in humans. It is likely that some of the physiological and psychological effects of caloric restriction that occur in animals may impact the human life very differently. For certain, calorie restriction has a plethora of health benefits in mammals, such as a reduction in age-related diseases such as cancer. However, despite the "magic" of CR, this intervention in humans may present itself with a number of health concerns, which may not be applicable to or impact the life of experimental animals, but may do so in humans. These potential pitfalls and "side effects" are not clearly addressed in the literature and will be a focus of this review.